



THE UNIVERSITY
of ADELAIDE

OFFICIAL ACADEMIC TRANSCRIPT

Name: Joshua Rikihana John Tomlin
Student ID: 1867443
Issue Date: 15 October 2025

Program Summary	Code	Status	GPA
Doctor of Philosophy	PHMCS	Conferred on 09 October 2025	Not Available

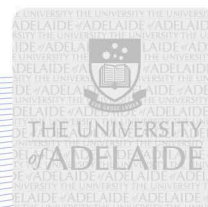
Candidature Summary	Units
2025 PHMCS Academic Area: Mathematical Sciences Thesis submitted on: 30 June 2025 Thesis Title: A Connected Sum Formula for the Seiberg-Witten Invariant of 4-Manifold Families	11.01
2024 PHMCS Academic Area: Mathematical Sciences	24.00
2023 PHMCS Academic Area: Mathematical Sciences School	21.26
2022 PHMCS Academic Area: Mathematical Sciences School Research Commencement Date: 24 January 2022 Approved Research Topic: Families Seiberg-Witten Invariants of 4-Manifold Families.	22.48

Candidature Achievements
2025 PHMCS Completed 120h Career and Research Skills Training

Prizes
2025 PHMCS Dean's Commendation for Doctoral Thesis Excellence

---END OF TRANSCRIPT---

Professor Peter Høj AC
Vice-Chancellor and President



THE UNIVERSITY OF ADELAIDE

Established in 1874

GRADES AND CORRESPONDING MARKS

Grade	Description
High Distinction	A mark between 85-100
Distinction	A mark between 75-84
Credit	A mark between 65-74
Pass	A mark between 50-64
Non-Graded Pass	Satisfactory performance
Fail	A mark between 1-49
Fail No Submission	No work submitted for assessment. This is counted as a fail
No Formal Examination	No Formal Examination
Continuing	Course taught over two or more semesters
Result Pending	An interim result
1	First Class Honours
2A	Second Class Div. A Honours
2B	Second Class Div. B Honours
3	Third Class Honours
Not Awarded	Not awarded Honours
Withheld	An interim result
Withdraw No Fail	Student withdrew before the designated deadline. This is not counted as a fail
Withdraw Fail	Student withdrew after the designated deadline. This is counted as a fail

This explanation of grades is effective from January 2011. For details of grades and notations in use prior to 2011 refer to <http://www.adelaide.edu.au>. This website also contains information regarding grading systems used for the former Roseworthy Agricultural College and for the South Australian College of Advanced Education which amalgamated with the University in 1991.

Grade Point Average (GPA)

A GPA is calculated for each coursework program in which a student is enrolled (with some exceptions). The GPA is displayed as a value between 0 and 7, based on the weighting of each course and the final grade achieved. The formula for determining a GPA is available at <http://www.adelaide.edu.au>

Northwestern University

Office of the Registrar – Transcript Services

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Evanston, IL 60208-1118

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<http://www.registrar.northwestern.edu>

26-Oct-2021

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Northwestern University
633 Clark Street
Evanston, IL 60208
United States

Name: Tomlin, Joshua Rikihana John
Student ID: 3257023

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Official Graduate Transcript

Print Date: 10/26/2021

Beginning of The Graduate School Record

2020 Fall (09/16/2020- 12/12/2020)

Program: Mathematics: PhD

Course	Description	Attempted	Earned	Grade	Points
MATH 430-1	Dynamical Systems	1.000	1.000	A	4.000
MATH 483-1	Algebraic Geometry	1.000	1.000	A	4.000
MATH 514-1	Topics in Geometry	1.000	1.000	A	4.000
MATH 517-1	Topics in Algebra	1.000	1.000	A	4.000

2021 Winter (01/11/2021- 03/20/2021)

Program: Mathematics: PhD

Course	Description	Attempted	Earned	Grade	Points
COMP_SCI 496-0	Special Topics in CS	1.000	1.000	A	4.000
Course Topic:	Advanced Topics in Modern Cryptography				
MATH 515-1	Topics in Geometry & Topology	1.000	1.000	A	4.000
MATH 518-1	Topics in Number Theory	1.000	1.000	A	4.000
MATH 520-1	Topics in Mathematical Physics	1.000	1.000	A	4.000

2021 Spring (03/30/2021- 06/12/2021)

Program: Mathematics: PhD

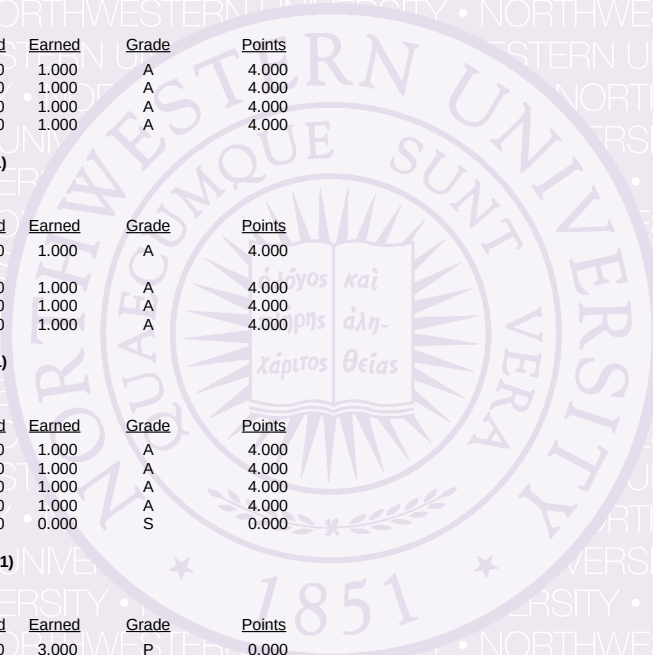
Course	Description	Attempted	Earned	Grade	Points
MATH 450-3	Probability Stochastic Analysis	1.000	1.000	A	4.000
MATH 483-2	Algebraic Geometry	1.000	1.000	A	4.000
MATH 514-2	Topics in Geometry	1.000	1.000	A	4.000
MATH 515-2	Topics in Geometry & Topology	1.000	1.000	A	4.000
MATH 519-0	RCR Training	0.000	0.000	S	0.000

2021 Summer (06/21/2021- 08/28/2021)

Program: Mathematics: PhD

Course	Description	Attempted	Earned	Grade	Points
MATH 590-0	Research	3.000	3.000	P	0.000

End of Official Graduate Transcript



NORTHWESTERN UNIVERSITY
Evanston, Illinois

ACCREDITATION

Northwestern University is accredited by the Higher Learning Commission (www.hlcommission.org). Other professional, college, school, and departmental accreditations are listed here:
http://www.registrar.northwestern.edu/academic_records/index.html
Northwestern University's CEEB code is 001739.

OFFICE OF RECORD

The Office of the Registrar, 633 Clark Street, Evanston, Illinois 60208, (847) 491-5234, fax: (847) 491-8458, www.registrar.northwestern.edu, issues transcripts of records for the following schools in the University. Such transcripts are a complete and chronological listing of all courses attempted in any of the schools listed.

Bienen School of Music	Kellogg School of Management
Dental School (closed 2001)	School of Communication (formerly Speech)
The Graduate School	School of Education and Social Policy
McCormick School of Engineering and Applied Science	Weinberg College of Arts and Sciences
Graduate Nursing School (closed 1990)	Medill School of Journalism, Media, Integrated Marketing Communications
Physician Assistant Program	Prosthetics-Orthotics (Master's program only)

The Office of the Dean/Director of each school/program listed below issues transcripts of records for these units. These transcripts must be requested separately and in addition to any transcripts from other Northwestern schools.

Northwestern Pritzker School of Law (312) 503-8464, www.law.northwestern.edu
Northwestern University in Qatar 974-4454-5072, www.qatar.northwestern.edu
Feinberg School of Medicine (312) 503-1369 www.feinberg.northwestern.edu
School of Professional Studies (312) 503-6950 www.sps.northwestern.edu
Physical Therapy (312) 908-8160 www.feinberg.northwestern.edu/sites/pthms/
Prosthetics-Orthotics (certificates) (312) 503-5700 www.nupoc.northwestern.edu/

ACADEMIC CALENDARS

Northwestern University offers programs on numerous calendars. Unless listed specifically below, the calendar on this transcript is a quarter system consisting of three quarters lasting approximately 10 weeks and one summer session lasting 10-11 weeks. Terms may include shorter sessions. The Executive MBA Program through the Kellogg School of Management is an exception with class meetings on designated weekends during terms corresponding to the quarter calendar.

Northwestern University in Qatar operates on a traditional semester calendar consisting of two terms each lasting 16 weeks and one summer term.

The Physician Assistant Program operates on a trimester calendar consisting of three terms each lasting 16 weeks.

The Physical Therapy Program operates on a semester calendar consisting of three terms each lasting 16 weeks.

The Prosthetics-Orthotics master's program uses a course unit system in which a 1 in the earned and attempted column = 1-unit course. For the purpose of transfer credit, one unit should be considered to be the equivalent of four quarter hours or 2 2/3 semester hours.

CREDIT

For quarter-based programs, in September 1969 NU began using a course unit system in which a 1 in the earned and attempted column = 1-unit course. For the purpose of transfer credit, one unit should be considered to be the equivalent of **four quarter hours or 2 2/3 semester hours**.

Prior to 2006 the Summer Session was based on a semester system and credits taken in that context should be considered to be the equivalent of four quarter hours or three semester hours.

For an explanation of credits earned in quarter-based programs prior to 1969:
http://www.registrar.northwestern.edu/academic_records/index.html

The Physician Assistant program uses a trimester hour credit measure in which each credit hour corresponds to an hour of meeting time for each week of a 16-week trimester.

The Physical Therapy program uses a semester hour credit measure in which each credit hour corresponds to an hour of meeting time for each week of a 16-week semester.

The Prosthetics-Orthotics master's program uses a course unit system in which a 1 in the earned and attempted column = 1 course. For the purpose of transfer credit, one course should be considered to be the equivalent of four quarter hours or 2 2/3 semester hours.

Northwestern University in Qatar uses a course unit system in which a 1 in the earned and attempted column = 1-unit course. For the purposes of transfer credit, one unit should be considered to be the equivalent of four semester hours.

GRADE POINT AVERAGE (GPA)

All courses attempted are recorded on the transcript and used in the GPA calculation. GPA is computed by taking the total grade points divided by the attempted units. NR, T, TR, P, N, K, S, U, and W grades are not included in GPA calculations.

Northwestern University does not calculate major GPAs nor does it rank its students.

EXPLANATION OF GRADE POINTS AND GRADES

For grade categories and years not represented below visit:
www.registrar.northwestern.edu/academic_records/index.html

Note: GPAs are not calculated on official graduate and professional transcripts.

ABC GRADING SCALE

Grade Points	Grade	Description
4.0	A	Excellent
3.7	A-	
3.3	B+	
3.0	B	Good
2.7	B-	
2.3	C+	
2.0	C	Satisfactory
1.7	C-	
1.0	D	Poor but passing
0.0	F	Fail
0.0	X	Missed final exam
0.0	Y	Work incomplete

BY SCHOOL, WHEN THE GRADE RUBRIC ABOVE IS APPLICABLE

Undergraduate Programs	September 1982 - present
Bienen School of Music (graduate programs)	January 2005 – present
School of Communication (graduate programs)	September 2005 - present
School of Education and Social Policy (graduate programs – D grade not used)	March 2005 - present
The Graduate School (D grade not used)	September 2004 - present
Medill School of Journalism, Media, Integrated Marketing Communications (graduate programs – D grade not used)	October 1986 - present
McCormick School of Engineering and Applied Science (graduate programs)	September 1996 - present

GRADE POINTS AND GRADES USED BY KELLOGG SCHOOL OF MANAGEMENT (non-executive MBA Programs)

Grade Points	Grade	Description
4.0	A	Excellent
3.0	B	Good
2.0	C	Satisfactory
1.0	D	Poor but passing
0.0	F	Fail
0.0	X	Missed final exam
0.0	Y	Work incomplete

TRANSCRIPT NOTATIONS AND ABBREVIATIONS CURRENTLY IN USE

HP	High Pass
K	Indicates work in progress°
LP	Low Pass
N	No grade, no credit°
NR	No grade Reported by Instructor
P	Pass with credit°
S	Satisfactory (non-credit course)
T	Transfer grade° (Spring Quarter 1969-70, full academic credit)
TR	Transfer grade° (Qatar Campus)
U	Unsatisfactory (non-credit course)
V	Visitor (auditor)
W	Withdrew - with permission
X	Absent from final examination°°
Y	Incomplete - Additional work required°°
°	Not included in either the quarterly or the cumulative grade point average
°°	Carries zero grade points and included in calculation of GPA. Both the quarterly and cumulative GPA are changed if a final grade is reported

TRANSCRIPT SYMBOLS

*	Grades received by special report
#	Duplication
##	Not applicable toward degree

For transcript notations and symbols not described here:
http://www.registrar.northwestern.edu/academic_records/index.html

DEGREES AWARDED

For a complete list of degrees awarded:
www.registrar.northwestern.edu/academic_records/index.html

TRANSFER CREDIT

Undergraduate records document articulated transfer credit by listing the institution of record and a T grade for each approved course. Grades for work transferred from another institution are not recorded. If such grades are needed the student must request a transcript directly from the awarding institution.

STATUS

Students should be regarded as in good academic standing unless otherwise noted. Each unit devises a probation/suspension/withdrawal policy, as well as academic eligibility to re-enroll after an absence.

COURSE NUMBERING SYSTEM

A/100 level	Courses primarily for freshmen and sophomores, usually without college prerequisite.
B/200 level	Courses primarily for sophomores and juniors, usually with the prerequisite of an A/100 level course in the same or a related department.
C/300 level	Courses primarily for upperclassmen and graduates, often with the prerequisites of an A/100 and/or B/200 level course in the same or a related department.
D/400 level	Courses or seminars primarily for graduates, in which the major part of the work is not research.
E/500 level	Courses for graduates only; seminars in which the work is primarily research, or special research by the individual student under faculty direction.

COURSE SUBJECTS AND DESCRIPTIONS

For more details: www.northwestern.edu/caesar/

This Transcript Key was last updated in September 2016.

TRANSCRIPT OF ACADEMIC RECORD

NAME

JOSHUA TOMLIN

STUDENT No.

5643046**AWARD AND PROGRAM SUMMARY**

DESCRIPTION	STATUS	DATE
BACHELOR OF STUDIES	DISCONTINUED	
BACHELOR OF PHILOSOPHY (HONOURS) (WITH FIRST CLASS HONOURS) SPECIALISATION IN MATHEMATICS HONOURS	AWARDED	10 DECEMBER 2019

ENROLMENT DETAILS

	COURSE CODE	COURSE TITLE	UNITS TAKEN	MARK	GRADE
UNDERGRADUATE GPA: 7					
3050	BACHELOR OF STUDIES				
2014	AUTUMN SESSION				
	EXTN1011A	ANU EXTENSION PHYSICS A	0		
2015	AUTUMN SESSION				
	EXTN1011B	ANU EXTENSION PHYSICS B	6	86	HD
4660	BACHELOR OF PHILOSOPHY (HONOURS)				
2016	FIRST SEMESTER				
	CHEM1101	CHEMISTRY 1 HONOURS PATHWAYS OPTION	6	85	HD
	COMP1130	INTRODUCTION TO PROGRAMMING AND ALGORITHMS (ADVANCED)	6	90	HD
	MATH1115	ADVANCED MATHEMATICS AND APPLICATIONS 1	6	90	HD
	PHYS1101	PHYSICS I ADVANCED STUDIES EXTENSION ASTROPHYSICS MOOC	6	93	HD
2016	SECOND SEMESTER				
	COMP1140	INTRODUCTION TO SOFTWARE SYSTEMS (ADVANCED)	6	100	HD
	MATH1116	ADVANCED MATHEMATICS AND APPLICATIONS 2 ADVANCED STUDIES EXTENSION MATHS COMMUNICATION (NUMBER THEORY)	6	93	HD
	MATH2322	ADVANCED ALGEBRA 1: GROUPS, RINGS AND LINEAR ALGEBRA ADVANCED STUDIES EXTENSION ALGEBRA 1 ASE	6	96	HD
	PHYS1201	PHYSICS 2 ADVANCED STUDIES EXTENSION ASTROPHYSICS MOOC	6	94	HD
2017	FIRST SEMESTER				
	MATH2320	ADVANCED ANALYSIS 1: METRIC SPACES AND APPLICATIONS ADVANCED STUDIES EXTENSION AXIOMATIC SET THEORY, JORDAN MEASURE, AND AN INTRODUCTION TO FOURIER SERIES	6	96	HD

Certified Correct (see reverse side for gradings)

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Ariel Edge
Registrar, Student Administration
10 December 2019

TRANSCRIPT OF ACADEMIC RECORD

NAME

JOSHUA TOMLIN

STUDENT No.

5643046

ENROLMENT DETAILS

	COURSE CODE	COURSE TITLE	UNITS TAKEN	MARK	GRADE
	MATH3345	ADVANCED ALGEBRA 2: FIELD EXTENSIONS AND GALOIS THEORY	6	92	HD
	PHYS2013	QUANTUM MECHANICS	6	92	HD
	PHYS2020	THERMAL AND STATISTICAL PHYSICS	6	88	HD
		HONOURS PATHWAYS OPTION			
	SCNC2101	ADVANCED STUDIES 1	6	82	HD
		ADVANCED STUDIES COURSE			
		REPRESENTATION THEORY			
2017	SECOND SEMESTER				
	MATH3228	ADVANCED COMPLEX ANALYSIS	6	94	HD
	MATH3354	ADVANCED TOPICS IN ALGEBRA	6	91	HD
	MATH4204	ALGEBRAIC TOPOLOGY HONOURS	6	81	HD
	PHYS2016	ELECTROMAGNETISM	6	90	HD
		ADVANCED STUDIES EXTENSION			
		ADVANCED QUESTIONS IN ELECTROMAGNETISM			
2018	FIRST SEMESTER				
	COMP4670	INTRODUCTION TO STATISTICAL MACHINE LEARNING	6	83	HD
	MATH3320	ADVANCED ANALYSIS 2: TOPOLOGY, LEBESGUE INTEGRATION AND HILBERT SPACES	6	90	HD
	MATH3342	ADVANCED DIFFERENTIAL GEOMETRY	6	97	HD
	MATH3349	SPECIAL TOPICS IN MATHEMATICS	6	90	HD
		MATHS TOPIC #1			
		THREE-MANIFOLDS			
	SCNC2102	ADVANCED STUDIES 2	6	85	HD
		SPECIAL TOPICS - CATEGORY THEORY			
2018	SECOND SEMESTER				
	MATH3325	ADVANCED FUNCTIONAL ANALYSIS, SPECTRAL THEORY AND APPLICATIONS	6	94	HD
	MATH3349	SPECIAL TOPICS IN MATHEMATICS	6	95	HD
		MATHS TOPIC #1			
		SPECIAL TOPICS ON GAUGE THEORY			
	MATH3354	ADVANCED TOPICS IN ALGEBRA	6	92	HD
	SCNC2103	ADVANCED STUDIES 3	6	88	HD
		ADVANCED STUDIES COURSE			
		SPECIAL TOPICS IN MATHEMATICS - INFINITY CATEGORIES			
2019	FIRST SEMESTER				
	MATH4005	THESIS AND PRESENTATIONS IN MATHEMATICS	6		KU *
	MATH4349	SPECIAL TOPICS IN MATHEMATICS	6	91	HD
		LINEAR LIE GROUPS AND LIE ALGEBRAS			
	MATH4349	SPECIAL TOPICS IN MATHEMATICS	6	96	HD
		EQUIVARIANT STABLE HOMOTOPY THEORY			

Certified Correct (see reverse side for gradings)

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Ariel Edge
Registrar, Student Administration
10 December 2019

TRANSCRIPT OF ACADEMIC RECORD

NAME

JOSHUA TOMLIN

STUDENT No.

5643046

ENROLMENT DETAILS

	COURSE CODE	COURSE TITLE	UNITS TAKEN	MARK	GRADE
	MATH6211	TOPICS IN MATHEMATICAL PHYSICS	6	97	HD
2019	SECOND SEMESTER				
	HONS4602	FINAL HONOURS GRADE	0	93	H1
	MATH4005	THESIS AND PRESENTATIONS IN MATHEMATICS	18	90	HD
	MATH4349	SPECIAL TOPICS IN MATHEMATICS	6	98	HD
		TOPIC 1			
		SYMPLECTIC GEOMETRY			

PRIZES AND SCHOLARSHIPS

2016 ANU NATIONAL UNIVERSITY SCHOLARSHIP

total number of courses printed on academic transcript: 35

Entries below this line are NOT VALID

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Ariel Edge
Registrar, Student Administration
10 December 2019

The Australian National University Transcript of Academic Record

Explanation of grades and codes

Key to grades and codes				
HD	High Distinction	80 - 100	DA	Deferred assessment
D	Distinction	70 - 79	IP	In progress (continuing course)
CR	Credit	60 - 69	KU	Continuing course
P	Pass	50 - 59	EE	Enrolled elsewhere
CRS	Course requirements satisfied	ungraded pass	EXE	Exemption external (no unit value)
PS	Pass at supplementary exam	50	EXI	Exemption internal (no unit value)
HLP	Higher level pass (Medical School)	higher ungraded pass	PX	Offered supplementary assessment
N	Fail	0 - 49	RC	Research continuing
NCN	Not completed/Fail	not graded	RP	Result pending
WD	Withdrawn without failure	not graded	STE	Status external (external credit)
WN	Withdrawn with failure	not graded	STI	Status internal (internal credit)
H1	First Class Honours	80 - 100	TS	Thesis submitted
H2A	Second Class Honours Division A	70 - 79	WA	Grade withheld for administrative reason
H2B	Second Class Honour Division B	60 - 69	WF	Grade withheld for non payment of fees
H3	Third Class Honours	50 - 59		

GPA, previous grades and unit values

ANU calculates GPA on a 7-point scale.

*Indicates a grade is not included in the GPA calculation. For further information including program exclusions, please go to anu.edu.au

The majority of courses offered at ANU are valued at 6 units. The normal maximum load in a semester or half-year is 24 units (normally 4 courses). Undertaking 18-24 units per half-year is considered a full-time study load. ANU considers 30 European Credit Transfer System (ECTS) credits as equivalent to 24 ANU units.

In 1994 a uniform grading system was introduced for all coursework students. Since 1994 the mark, as well as the grade obtained, has been disclosed on the Transcript of Academic Record. Details of grades and codes prior to 1994, prior unit value equivalencies and unit requirements for degree programs are available at anu.edu.au

Program status

Status codes	
Active	Currently holds an active place in the program
Completed	Satisfied academic requirements for award
Awarded	Award conferred
Discontinued	Ceased enrolment prior to the completion of the program (includes internal program transfers)
Excluded	Excluded from the program

Semesters and sessions

Coursework programs		Research programs	
First Semester	February - June	First Semester	1 January - 30 June
Second Semester	July - November	Second Semester	1 July - 31 December
Summer Session	January - March	Quarter 1	January - March
Autumn Session	April - June	Quarter 2	April - June
Winter Session	July - September	Quarter 3	July - September
Spring Session	October - December	Quarter 4	October - December

Other information

Details of programs and courses may be found at programsandcourses.anu.edu.au

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These explanations are authorised by the Registrar and are current as of 1 January 2015. Subsequent amendments are available at anu.edu.au